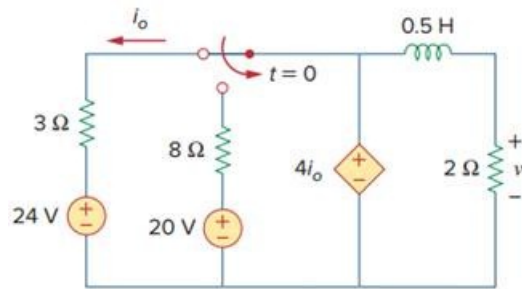


Problem

Find $v(t)$ for $t < 0$ and $t > 0$ in the circuit of Fig. 7.121.

Fig. 7.121:



Step-by-step solution

Step 1 of 9

Refer to figure 7.121 given in the textbook.

As all the independent sources are DC sources, short circuit the inductor initially.

Redraw the circuit for $t < 0$ as shown in figure 1.

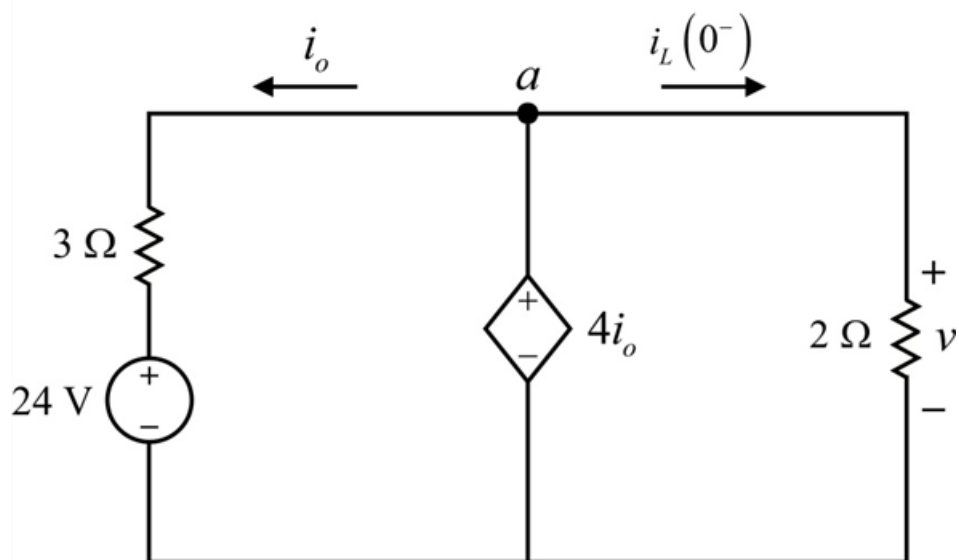


Figure 1

Step 2 of 9

Apply Kirchhoff's voltage law to the circuit shown in figure 1.

$$4i_o - 3i_o - 24 = 0$$

$$i_o = 24 \text{ A}$$

Step 3 of 9

Calculate voltage v .

Consider the following for v .

$$v = 4i_o$$

Substitute 24 for i_o .

$$\begin{aligned} v &= 4 \times 24 \\ &= 96 \text{ V} \end{aligned}$$

Therefore, the voltage $v(t)$ for $t < 0$ is 96 V.

Step 4 of 9

Calculate the current flowing through the inductor for $t < 0$.

$$i_L(0^-) = \frac{v}{2}$$

Here, $i_L(0^-)$ is the current flowing through the inductor for $t < 0$.

Substitute 96 for v .

$$\begin{aligned} i_L(0^-) &= \frac{96}{2} \\ &= 48 \text{ A} \end{aligned}$$

Consider the following equation:

$$i_L(0^-) = i_L(0^+) = i(0) = 48 \text{ A}$$

Here, $i_L(0^+)$ is the current flowing through the inductor for $t > 0$.

Step 5 of 9

Short circuit the dependent voltage source of $4i_o$, because the value of current i_o becomes zero for $t > 0$.

Draw the circuit for $t > 0$ as shown in figure 2.

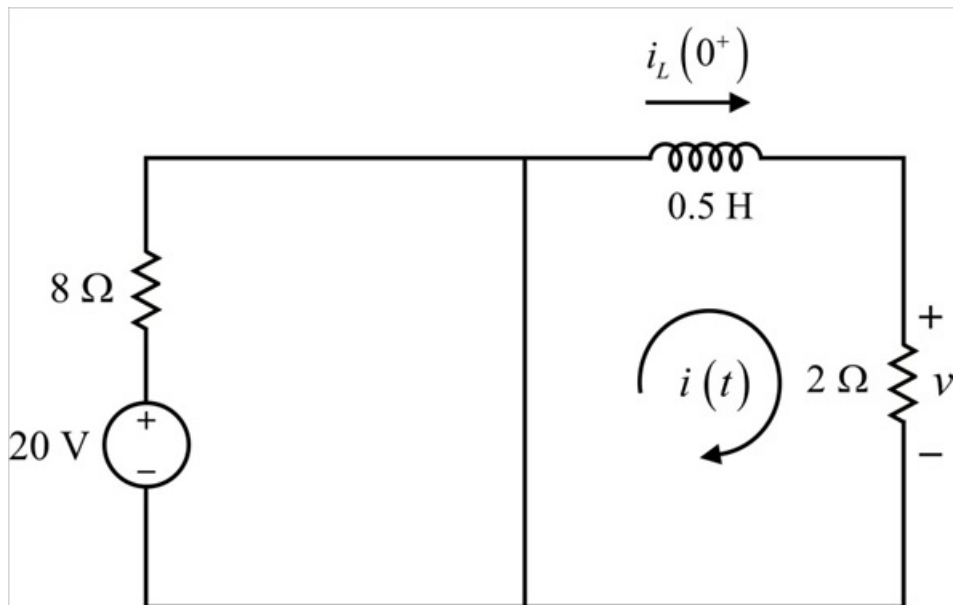


Figure 2

Step 6 of 9

Determine current $i(t)$.

$$i(t) = i(\infty) + [i(0) - i(\infty)]e^{\frac{-t}{\tau}}$$

As there is no source connected, the steady state value $i(\infty)$ becomes zero.

Substitute 48 for $i(0)$, and 0 for $i(\infty)$.

$$i(t) = 0 + [48 - 0]e^{\frac{-t}{\tau}}$$

$$i(t) = 48e^{\frac{-t}{\tau}} \text{ A} \dots\dots (1)$$

Step 7 of 9

Calculate time constant τ .

$$\tau = \frac{L}{R}$$

Substitute 0.5 for L , and 2 for R .

$$\begin{aligned}\tau &= \frac{0.5}{2} \\ &= \frac{1}{4}\end{aligned}$$

Step 8 of 9

Recall equation (1).

$$i(t) = 48e^{\frac{-t}{\tau}} \text{ A}$$

Substitute $\frac{1}{4}$ for τ .

$$\begin{aligned}i(t) &= 48e^{\left(\frac{-t}{\left(\frac{1}{4}\right)}\right)} \text{ A} \\ &= 48e^{-4t} \text{ A}\end{aligned}$$

Step 9 of 9

Determine the voltage $v(t)$ for $t > 0$.

$$v(t) = 2i(t)$$

Substitute $48e^{-4t}$ for $i(t)$.

$$\begin{aligned}v(t) &= 2 \times 48e^{-4t} \\ &= 96e^{-4t} \text{ V}\end{aligned}$$

Therefore, the voltage $v(t)$ is $\boxed{96e^{-4t} \text{ V}}$.